Optimization of Deep Neural Network based Hand Gesture Classification Model for Large-Scale Dataset

Jie Yu[1], Tao Xu[1,2,*], Zhiquan Feng[1,2,*], Weifeng Wang[1], Li Ma[1] and Xinyi Diao[1]

1 School of Information Science and Engineering, University of Jinan, Jinan 250022, China

2 Shandong Provincial Key Laboratory of Network Based Intelligent Computing, Jinan 250022, China

xutao@ujn.edu.cn; ise_fengzq@ujn.edu.cn

ABSTRACT

Hand gesture classification is a key step of gesture based human-computer interaction. The hand gesture dataset with depth information is increasing faster which brings challenges for deep neural network-based hand gesture classification model, such as data loading of big gesture data, optimization of model and visualization of training stage. In this paper, an optimization method of deep neural network-based hand gesture classification model for large-scale dataset is proposed, which improves the efficiency of data loading and makes the model have better time complexity and space complexity. Experimental results show that the proposed optimization method can effectively improve the training efficiency of classification model for large gesture data sets, and make classification models easy to train in a unified framework, which provides a good foundation for the research of hand gesture classification model.

CCS Concepts

Human-centered computing~Gestural input

Keywords

Deep neural network; optimization of model; hand gesture recognition; large-scale dataset.

1. INTRODUCTION

Hands have a high degree of freedom, which gives the ability for human hands to be used in human-computer interaction. Gesture-based interaction is an active field of human-computer interaction, in which gesture recognition is the key step. Hand gesture recognition includes collecting gesture samples, data pre-processing, model training and gesture classification. In recent years, deep neural network has achieved good result in image classification. Many deep neural network-based hand gesture recognition algorithms have been proposed. Due to the wide popularity of Kinect and other depth supported devices, depth image can be easily achieved. Depth based hand gesture datasets make deep neural network-based model be very popular in the research area of hand gesture recognition.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

ICBDT2019, August 28–30, 2019, Jinan, China

© 2019 Association for Computing Machinery.

ACM ISBN 978-1-4503-7192-6/19/08…$15.00

https://doi.org/10.1145/3358528.3358571

However, with the increase of the scale of the hand gesture dataset, the existing gesture recognition methods have many problems to be solved, such as the loading of numerous samples, the validity check in the stage of model training, optimization and visualization of the architecture of deep neural network-based model. These problems bring challenges for training, parameter optimization, experimental comparison of hand gesture recognition models and recurrence of experimental results.

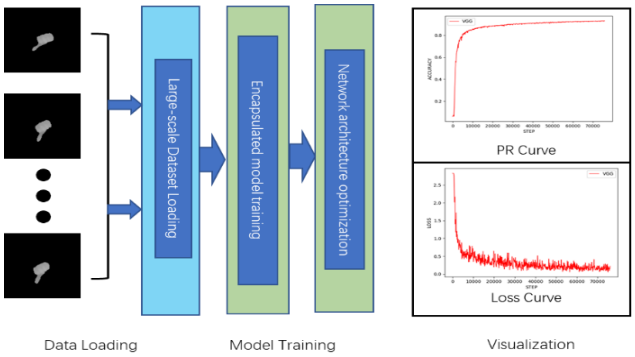

Figure 1. Framework of the proposed method.

Aiming at the above problems, this paper proposes an optimization process of gesture classification model based on deep neural network for large-scale gesture dataset. First, the flowchart of data loading is optimized, which involves data validity, data loading and dataset structure standardization. Second, an encapsulated generalized unified framework for model training and visualization of experimental results is presented in this paper. The unified framework includes an encapsulated unified interface for model training and classification, quantitative visualization of experimental results, etc. The proposed optimization method also involves efficiency optimization and spatial complexity optimization of deep neural network models. Experiment results show that the proposed method could effectively improve the training efficiency of deep neural network-based classification models for large-scale gesture dataset, and make the gesture classification models be easily trained, used and compared in a unified framework, which provides a good foundation for the research of classification model.

The rest of this paper is organized as follows: section 2 introduces the related work. Section 3 describes the data loading method for large-scale dataset during the processing of the hand gesture recognition, the optimization of architecture and the visualization of experimental results. The experimental results and analysis are given in section 4 and section 5 presents the conclusions.

2. RELATED WORK

Previous work on hand gesture recognition often utilized RGB based hand gesture images for model training. RGB images can be easily affected by illumination change which resulted in inaccurate training of gesture classification model based on RGB images. Depth information is robust to illumination change. Therefore, more and more researchers focus on depth images and proposed many depth-based gesture datasets and depth-based gesture recognition algorithms. Some commonly used hand gesture datasets are shown in Table 1.

Table 1. Hand gesture datasets

Name	Number of gestures	Samples of each gestures	Number of samples
MSRA	**9**	**17000(±20)**	**152903**
NYU	**3**	**72757(±50)**	**729085**
ICVL	**9**	**22056(±50)**	**330994**
UJN-3M-HANDS	**17**	**100000 (±5000)**	**3000000**

Datasets listed in Table 1. have been utilized for gesture recognition model training, and researches have proposed many algorithms. Vuletic et al. reported the results of a systematic review undertaken to identify characteristics of touchless/in-air hand gestures used in interaction interfaces [1]. Cheok et al. presented a review of hand gesture and sign language recognition techniques [2]. Ref. [3] introduced a set of meaningful shape features for hand gesture recognition and evaluated their method on NTU dataset based on linear discriminant analysis with state-of-the-art methods. They also prepared a comprehensive dataset (SBU-1) for different hand gestures containing 2170 images. In Ref. [4], a depth-based modification to the shape matrix is devised to describe hand silhouettes. Postures are recognized using depth matrix and 1-nearest neighbor strategy based on a dynamic gesture dataset. The proposed dataset contains 10 distinct gestures captured via Kinect. The commonly used gesture datasets, such as MSRA, NTU and ICVL were introduced in Ref. [5] through Ref. [6], and deep neural network-based model was also utilized for hand gesture recognition [7]. Other data-driven method [8] and neural network-based algorithm with novelty gesture dataset [9,10] have also achieved good gesture recognition performance.

Although the mentioned methods demonstrated good experimental results on specific datasets, there are still challenges in big dataset based algorithm optimization. The diversity of samples in existing datasets is insufficient, which bring challenges in improving the gesture recognition rate of deep neural network-based methods. For large-scale gesture datasets, i.e., UJN-3M-HANDS, issues such as efficient data loading, optimization of model training and visualization of experimental results must be addressed.

3. OPTIMIZATION OF ARCHITECHTURE

3.1 UJN-3M-HANDS dataset

Large-scale dataset is necessary for training of deep neural network-based hand gesture classification models. According to the requirements of human-interaction in virtual reality-oriented teaching environment, a large-scale depth-based hand gesture dataset was proposed. The dataset, named UJN-3M-HANDS, contained more than 3 million depth gesture samples.

There are 17 static gestures and 34 dynamic gestures in UJN-3M-HANDS. Some static gesture samples were demonstrated in Figure. 2. Compared with other data sets, UJN-3M-HANDS proposed in this paper is designed for specific teaching application scenarios, and the categories of gestures are consistent with those used in experimental teaching. In the stage of gesture collection, the distance between the camera and the collector is set to more than 2 meters. The advantage of UJN-3M-HANDS is that the gesture samples were acquired from 2000 individuals, while there are many replication samples in other datasets. The contributions presented in this paper are achieved based on the large-scale UJN-3M-HANDS.

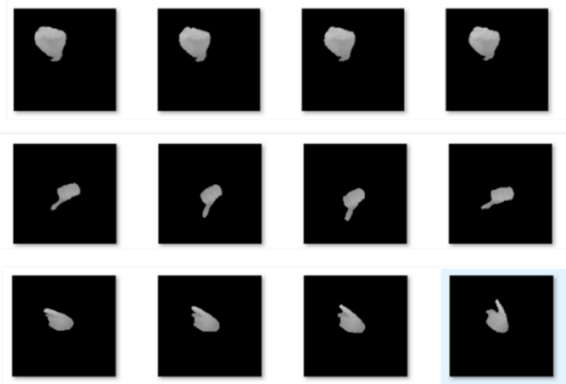

Figure 2. Hand gesture samples of UJN-3M-HANDS.

3.2 Efficient large-scale data loading

The existing gesture methods usually performed data loading in a variety of ways. Most of the data loading methods were not mentioned in previous work. The quantity of datasets utilized in training stage were small, which make researchers subjectively consider that data loading will not have a great impact on model training. However, the training requirements of large-scale datasets and deep neural network-based models make the data loading one of the key issues that researchers must consider.

In this section, an efficient data loading method for large-scale dataset was presented, and the flowchart is showed in Figure. 3.

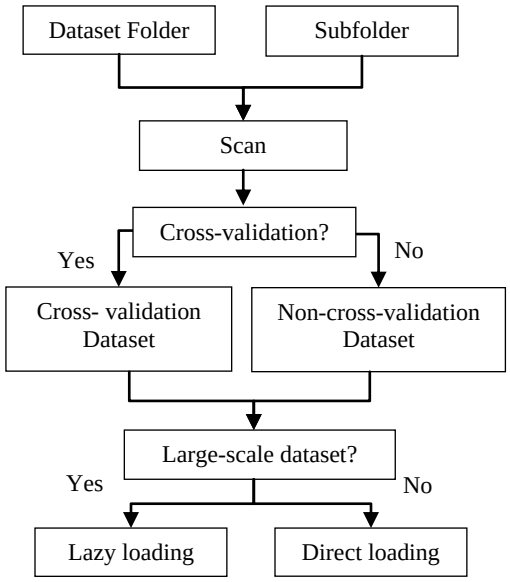

Figure 3. Flowchart of data loading.

When the dataset is loaded, the folder and subfolders are automatically scanned according to the configuration file, and dataset samples information can be obtained automatically. The data loading method could also generate non-cross-validation dataset and cross-validation dataset which could facilitate the training of models via a unified data loading framework. Moreover, considering the limited memory, the proposed data loading method provides two working modes for fast and efficient data loading of model training. One is a direct loading mode for small dataset, the other is a lazy loading mode for large-scale dataset. The efficient data loading method proposed in this paper has a strong generalization ability, and can theoretically adapt to the data loading task of all CNN based models' training.

3.3 Encapsulation of model training and experimental result visualization

The existing gesture recognition algorithms using different deep neural network architectures. There are various kinds of visual representations of experimental results, which brings difficulties for model training, reproduction of experimental results and comparisons with related methods. A generalized and unified encapsulation of CNN based framework and visualization of experimental results framework is presented in this section to solve these problems.

The framework provides function of inserting CNN based model into the unified framework as a plugin, and simplifies the training process of the CNN. Researchers only need to pass parameter and execute the entry function to train a CNN model. Lenet-5, Alexnet, VGG, and GoogNet are encapsulated in this framework and can be utilized directly, which makes the training of CNN based model much more convenient and the application of CNN models much easier.

For demonstration of experimental results, researchers use recognition rate with table, curves of loss function or precision-recall curve to denote the validity of proposed methods, which result in incomparability between various experimental results. Using the method which contains visualization and analysis of experimental results proposed in this paper, the above problems can be easily solved. This framework supports tracking the training data during the training process, saving the original data of key notes in training process. The proposed method also supports the simultaneous drawing of multiple groups of data and draw them together for convenient comparison.

3.4 Optimization of architecture

There are challenges for training a gesture recognition model based on large-scale dataset. For example, the training time of the model is very expensive due to high memory usage and improper coordination or repeated use of some optimization methods. Few people take this problem into considering because a lot of existing experiment use small dataset to train models. In this section, an optimization of architecture is proposed to improve the efficiency of gesture recognition models.

Figure. 4 illustrates the optimization method. Based on data loading method proposed in Section 3.2, the proposed optimization method firstly loads more than 3 million gesture samples into a server station with 8GB RAM. GPU devices can be found and used automatically without any extra settings in this framework.

One of the major contributions of optimization is removing Softmax layer behind the fully connected layer. Softmax formula is displayed in Equation 1. The meaning of the Softmax formula is when one z^j is bigger than the rest of the $\mathbf{z}$, the mapping component of z^j will be set to 1, and the other components will be set to 0.

$$\sigma(z)_j = \frac{e^{z_j}}{\sum_{k=1}^{K} e^{z_k}} \tag{1}$$

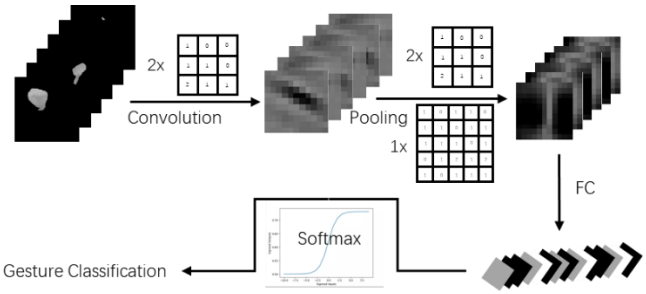

Figure 4. The optimization of CNN based model.

According to the analysis of Softmax, the optimization method remove the Softmax function followed by full connection layer (FC), while directly use value of neurons for classification. Another contribution of optimization is that the pooling operation is divided into two pooling parts, i.e., two 3x3 pooling operations. This optimization can ensure that the receptive field is maintained while reducing the network parameters of CNN. Batch normalization and dropout have similar effect, so only the most suitable one will be selected in A CNN based network.

4. EXPERIMENTAL RESULTS AND ANALYSIS

For evaluation of the proposed method, CNN based gesture recognition methods were trained and tested based on UJN-3M-HANDS. The experimental settings are as follows: CPU is 2x Intel Xeon E5-2620 with 64G RAM. GPU is a NVIDIA Tesla K40M with CUDA 9.0 and Pytorch 1.0.

1.4 million static gesture samples were used in the experiment, and two training modes, cross-validation and non-cross-validation were used to demonstrate the validity of the loading mode and lazy loading. The successful loading of data ensures the verification of subsequent optimization.

Lenet-5, Alexnet, VGG and Googlenet are utilized for gesture recognition. Table 2 presents the experimental results with optimization and without optimization.

Table 2. The accuracy of gesture recognition.

	Without optimization	With optimization
Lenet-5	**86.9%**	**88.4%**
AlexNet	**89.1%**	**90.7%**
GoogLenet	**88.9%**	**89.7%**
VGG	**92.5%**	**93.2%**

As showed in Table 2, CNN based gesture recognition methods achieved better accuracy with optimization proposed in this paper, which proves the effectiveness of the proposed optimization

method. For validity of efficiency of data loading method and encapsulation of model training and visualization of experimental results, the curves of loss function of CNN based methods were presented, and Figure 5 shows the loss curves with iteration steps.

In Figure 5, the loss function curves of Lenet-5 and VGG demonstrate the effectiveness of data loading, the efficiency of optimization process and the effectiveness of visualization of experimental results.

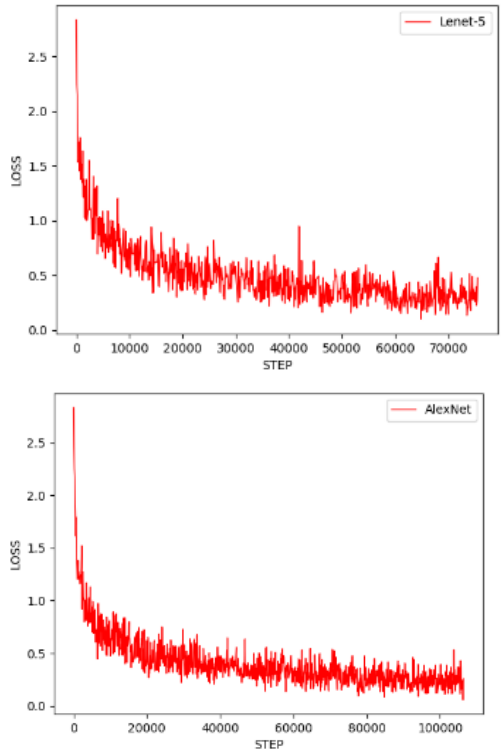

Figure 5. The curves of loss fucntion.

To verify the improvement of model training performance via the proposed optimization method, the training time, recognition rate(accuracy) and memory usage are used for quantitative comparison between Lenet-5 and VGG. Table 3 shows the quantitative results and Figure 6 shows the accuracy curves.

Table 3. The experimental results.

Item	Without optimization	After optimization
Training time (Lenet-5)	**29220s**	**28995s**
Accuracy (Lenet-5)	**0.7238**	**0.8843**
Memory usage (Lenet-5)	**21.52GB(RAM)**	**5.06GB(RAM)**
Training time (VGG)	**745991s**	**657474s**
Accuracy (VGG)	**0.9230**	**0.9315**
Memory usage (VGG)	**55.81GB(RAM) 21.02GB(SWAP)**	**7.16GB(RAM)**

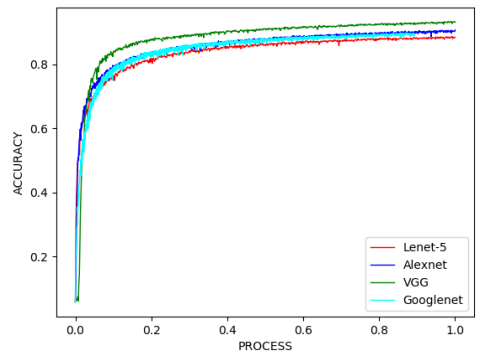

Figure 6. The result of CNNs after optimization.

5. CONCLUSIONS

An optimization of Deep Neural Network based Hand Gesture Classification Model for Large-Scale Gesture Dataset is proposed in this paper, including effective loading of large data sets, encapsulated model training and result visualization and deep neural network architecture optimization. The experimental results verify the effectiveness of the proposed optimization method, which is convenient and efficient for the training of deep neural network model on large-scale dataset.

6. ACKNOWLEDGMENTS

This work is supported by the National Key R&D Program of China (No. 2018YFB1004901), the Shandong Provincial Key R&D Program (No. 2017GGX10146), Doctoral Research Foundation of University of Jinan (No. XBS1653) and Natural Science Foundation of University of Jinan (No. XKY1804, XKY1928, XKY1931).

7. REFERENCES

[1] Vuletic T, Duffy A H, Hay L, et al. Systematic literature review of hand gestures used in human computer interaction interfaces[J]. International Journal of Man-machine Studies, 2019: 74-94.

[2] Cheok M J, Omar Z, Jaward M H, et al. A review of hand gesture and sign language recognition techniques[J]. International Journal of Machine Learning and Cybernetics, 2019, 10(1): 131-153.

[3] Mirehi N, Tahmasbi M, Targhi A T, et al. Hand gesture recognition using topological features[J]. Multimedia Tools and Applications, 2019: 1-26.

[4] Kane L, Khanna P. Depth matrix and adaptive Bayes classifier based dynamic hand gesture recognition[J]. Pattern Recognition Letters, 2019: 24-30.

[5] Qian C, Sun X, Wei Y, et al. Realtime and Robust Hand Tracking from Depth[C]. computer vision and pattern recognition, 2014: 1106-1113.

[6] Ren Z, Yuan J, Zhang Z, et al. Robust hand gesture recognition based on finger-earth mover's distance with a commodity depth camera[C]. acm multimedia, 2011: 1093-1096.

[7] Tompson J, Stein M, Lecun Y, et al. Real-Time Continuous Pose Recovery of Human Hands Using Convolutional Networks[J]. ACM Transactions on Graphics, 2014, 33(5):1-10.

[8] Tang D, Chang H J, Tejani A, et al. Latent Regression Forest: Structured Estimation of 3D Hand Poses[J]. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2017, 39(7): 1374-1387.

[9] Patil A R, Subbaraman S. A spatiotemporal approach for vision-based hand gesture recognition using Hough transform and neural network[J]. Signal, Image and Video Processing, 2019, 13(2): 413-421.

[10] Zengeler N, Kopinski T, Handmann U, et al. Hand Gesture Recognition in Automotive Human–Machine Interaction Using Depth Cameras[J]. Sensors, 2018, 19(1):1-27.